Experimental Analysis of Public Policies

Session 4

Compliance, IV, LATE, and External Validity

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Session Objectives

By the end of this session, you should be able to:

- explain how an instrument provides variation in an endogenous treatment;
- distinguish assignment Z from treatment received D;
- calculate an ITT, a first stage, and a Wald estimate;
- connect the Wald ratio to the general IV approach and 2SLS equations;
- use DAGs and behavioural schemas to assess IV assumptions;
- interpret the local effect and population in an empirical paper.

Roadmap

From Selection to Instrumental Variables

From Assignment Effects to the Wald Estimate

Assumptions and the Complier Effect

IV Evidence and Policy Interpretation

Connecting the Design to the General IV Approach

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From RCT Design to a Policy Question

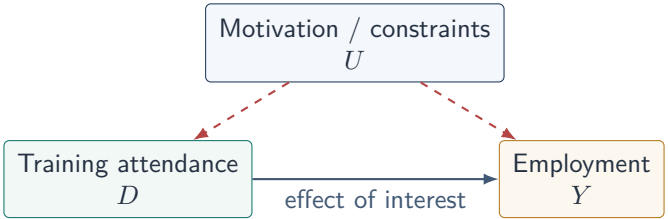
A job centre randomly invites eligible workers to training.

- Some invited workers do not attend.
- Some workers attend without an invitation.
- The city wants to know whether training improves employment.

Opening poll

Which comparison would you use: invited versus non-invited workers, or attendees versus non-attendees? What does each comparison answer?

Why Attendance Can Remain Endogenous

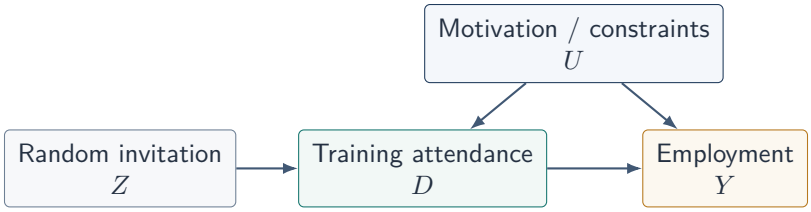


- Attendees may have different employment prospects even without training.
- Regression or matching needs the relevant confounders to be observed.

The missing counterfactual

How would the same workers have fared without training?

The IV Idea: Find an External Source of Treatment Variation

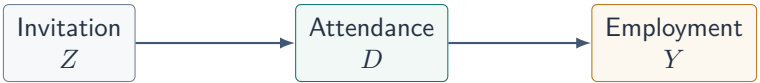


Instrumental variables (IV)

Use the variation in treatment generated by an external source Z.
Its causal interpretation depends on how Z is assigned and how it affects Y.

Here the random invitation is a **candidate instrument** for attendance.

Assignment Versus Receipt: Two Policy Questions



Effect of offering training

- Compare outcomes by random invitation.
- The city controls this policy lever.

Effect of attending training

- Attendance remains a choice.
- IV needs additional assumptions.

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Numerical Example: Calculate Before Naming

Illustrative training experiment: outcomes are measured after one year.

	Not invited $Z = 0$	Invited $Z = 1$
Attend training	20%	60%
Employed	45%	53%

Two minutes in pairs

Calculate the invitation's effect on attendance and on employment.
Which number directly describes the effect of the city's offer?

ITT and First Stage: Equations and Interpretation

First stage: invitation → attendance

$$ITT_D = E[D_i \mid Z_i = 1] - E[D_i \mid Z_i = 0]$$

$$\widehat{ITT}_D = 0.60 - 0.20 = 0.40$$

The offer raises attendance by **40 percentage points**.

ITT: invitation → employment

$$ITT_Y = E[Y_i \mid Z_i = 1] - E[Y_i \mid Z_i = 0]$$

$$\widehat{ITT}_Y = 0.53 - 0.45 = 0.08$$

The offer raises employment by **8 percentage points**.

A policy effect in its own right

The ITT measures the effect of offering the programme as implemented.
Without hats: population effects. With hats: sample estimates.

The Wald Estimate: Outcome Change per Induced Treatment

$$\widehat{\tau}_{IV} = \frac{\bar{Y}_{Z=1} - \bar{Y}_{Z=0}}{\bar{D}_{Z=1} - \bar{D}_{Z=0}} = \frac{\widehat{ITT}_Y}{\widehat{ITT}_D} = \frac{0.08}{0.40} = 0.20$$

- The offer changes attendance for only part of the population.
- The ratio rescales the outcome effect by the change in attendance.

A conditional interpretation

Under the IV assumptions, the estimated employment effect is **20 percentage points for workers induced to attend by the invitation**.

Roadmap

From Selection to Instrumental Variables

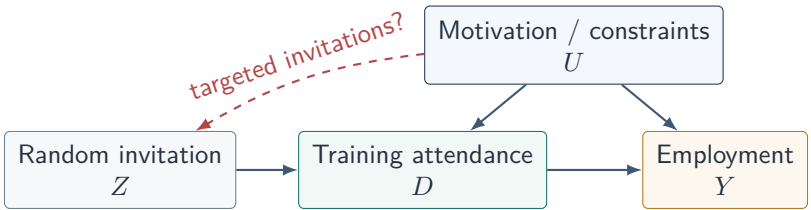
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Independence: Who Receives the Instrument?

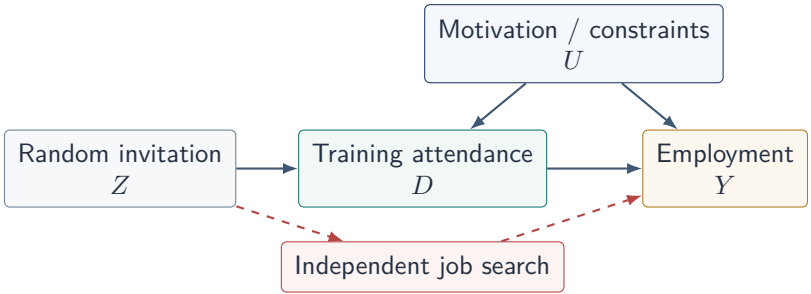


As good as randomly assigned

Z must be independent of potential outcomes and potential treatment responses (possibly conditional on pre-treatment controls X).

If motivated workers receive more invitations, the red path undermines the design.

Exclusion: Can the Instrument Bypass Treatment?



Exclusion restriction

The instrument affects the outcome only through treatment received.

A motivational letter could change job search even without attendance.
Random assignment still identifies the **offer effect**; this bypass threatens the IV interpretation.

Compliance Types: The Same Person in Two Worlds



Potential responses, not observed labels

We observe one assignment per person. Individual compliance types are generally unobserved.

Monotonicity: Which Behaviour Are We Ruling Out?



No defiers

The invitation does not reduce anyone's treatment receipt:

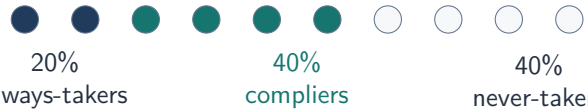
$$D_i(1) \geq D_i(0) \quad \text{for every person.}$$

Discuss

Could an invitation make someone suspicious enough to cancel their planned attendance?

Why Divide by 0.40? A Population Picture

Illustrative population: random assignment and no defiers.



- Without an invitation: only always-takers attend (20%).
- With an invitation: always-takers and compliers attend (60%).
- Under exclusion, assignment changes outcomes only for compliers.

$$\underbrace{0.08}_{\text{offer effect}} = \underbrace{0.40}_{\text{complier share}} \times \underbrace{0.20}_{\text{average effect for compliers}}$$

Local Average Treatment Effect

LATE: the average treatment effect for people induced to participate by Z. Under independence, exclusion, relevance and monotonicity:

$$ITT_D = \Pr(\text{complier}),$$
$$ITT_Y = \Pr(\text{complier}) \times LATE.$$

$$\Rightarrow \quad LATE = \frac{ITT_Y}{ITT_D} = E[Y_i(1) - Y_i(0) \mid D_i(1) > D_i(0)].$$

- LATE is local to a behavioural margin.
- It may differ from ATE or ATT.

IV Assumptions: Debrief Grid

Assumption	Question for the empirical setting
Relevance	Does the instrument visibly move treatment received?
Independence	Who or what determines the instrument?
Exclusion	Could the instrument affect outcomes without treatment?
Monotonicity	Is it plausible that nobody moves opposite to the instrument?

Most common exam mistake

Reporting a Wald ratio without saying which assumption is most fragile.

Design-Threat Lab: Which Assumption?

1. Letters are sent mainly to workers judged most employable.
2. Random letters include job-search advice useful without training.
3. Random letters change attendance by almost nothing.

Three minutes

For each case, name the threat and draw the relevant arrow or comparison.
Which case leaves a credible offer effect but threatens the IV treatment effect?

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From Random Encouragement to Institutional Instruments

Randomised encouragement

- A researcher or policymaker randomises an offer.
- Actual receipt remains a choice.

Institutional variation

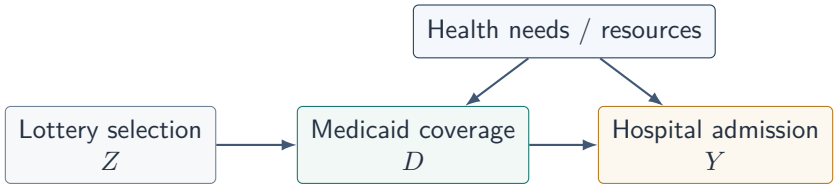
- An existing lottery or rule shifts treatment.
- Its assignment mechanism needs scrutiny.

> **The general IV approach**
> Find a credible source Z of variation in an endogenous treatment D, then defend the assumptions linking that variation to outcomes Y.

Two applications: health insurance in Oregon and the US draft lottery.

Oregon: Does Health Insurance Change Access to Care?

Problem: people with and without insurance differ in health and resources.



In 2008, selection offered uninsured low-income adults a chance to apply.
Some selected applicants did not obtain coverage; some controls became covered.

Finkelstein et al. (2012), QJE. Comparisons account for household size.

Oregon: Read the Three Estimates Together

Quantity	Estimate (pp)	SE (pp)
First stage: ever on Medicaid	+25.6	0.35
ITT: any hospital admission	+0.54	0.19
IV: any hospital admission	+2.1	0.74

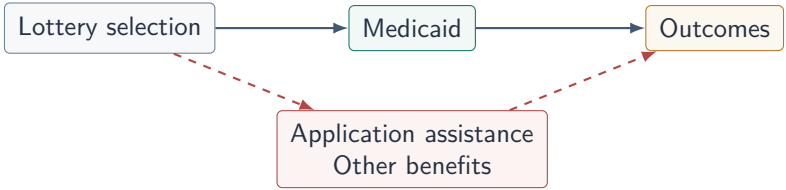
$$\frac{0.54 \text{ pp}}{0.256} \simeq 2.1 \text{ pp}$$

Interpret before judging

For whom does the 2.1-point estimate apply?
Does increased hospital use by itself establish improved health?

Finkelstein et al. (2012), Tables III–IV. Rounded adjusted estimates.

Oregon: A Literature-Based Exclusion Question



- Selection could help applicants access other assistance, bypassing Medicaid.
- The authors discuss this issue and find small effects on food-stamp receipt.

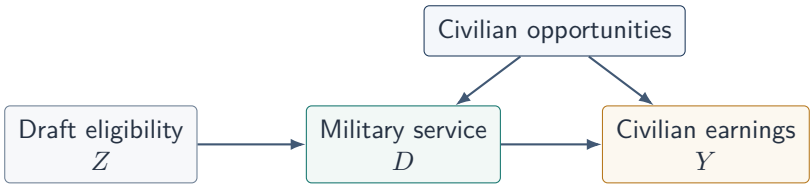
Design judgement

Why does a strong coverage first stage leave this question open?

Finkelstein et al. (2012), p. 1081: exclusion restriction discussion.

Draft Lottery: Does Military Service Affect Later Earnings?

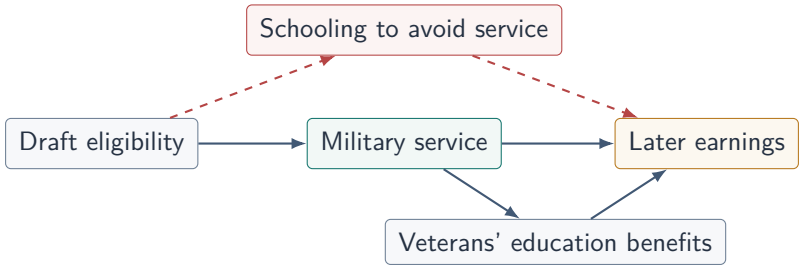
Problem: civilian opportunities influence who volunteers for military service.



Lottery numbers relative to a call-up ceiling shifted the probability of service. Angrist estimates an early-1980s earnings penalty of about 15% for white men whose service is shifted by this instrument, under IV assumptions.

Angrist (1990), AER, pp. 313–315 and 330.

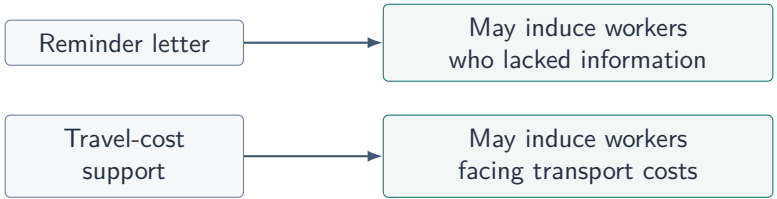
Draft Lottery: A Mechanism or an Exclusion Violation?



Bypass: eligibility changes schooling without service — exclusion concern.
Mechanism: service changes education through veterans' benefits — part of its total effect.

Angrist (1990), p. 330: draft-avoidance schooling discussion.

External Validity: A Different Instrument Can Move Different People



- Each IV estimate describes its own instrument-induced treatment margin.
- Even valid instruments can identify different effects if effects vary.

Policy question

Does this study move the same people, into the same treatment, as our proposed policy?

Weak First Stages and Uncertainty

A well measured denominator

$$0.08/0.40 = 0.20$$

A small, noisy denominator

$$0.01/0.02 = 0.50$$
$$0.01/0.01 = 1.00$$

- A poorly measured first stage makes the ratio unstable.
- Conventional IV confidence intervals can be unreliable with weak instruments.
- Use appropriate IV inference; account for clustering in assignment.

Two separate questions

Strength asks whether Z moves D precisely enough.
Validity asks whether the instrument identifies a causal treatment effect.

Scale-Up

- Treatment quality can change when a programme expands.
- Participants at scale may differ from pilot participants.
- General equilibrium effects may appear.
- Administrative capacity can become a binding constraint.

Policy example: a voucher may work for parents with nearby childcare slots but not after expansion if slots become scarce.

R Mini-Lab: From the Policy Offer to a Local Effect

Question: in a simulated encouragement design, what does the offer change?

```
# After running r_exercises/week_04_iv_late.R
m <- aggregate(cbind(Y, D) ~ Z, data = dat, mean)
itt <- m$Y[m$Z == 1] - m$Y[m$Z == 0]
first_stage <- m$D[m$Z == 1] - m$D[m$Z == 0]
c(ITT = itt, first_stage = first_stage,
  Wald = itt / first_stage)
```

Five minutes: compute, then interpret

Write one sentence about the offer and one about the induced recipients.
Here Y is a continuous outcome score: the Wald estimate is in **score units**.

Reading an IV Paper: Three Causal Questions

1. **Where does the identifying variation come from?**
   Name Z, D, Y; read the first stage and reduced form.
2. **What makes the comparison causal?**
   Draw the DAG and defend the most fragile assumption.
3. **Whose effect does the estimate describe?**
   Identify the treatment margin, study population and policy relevance.

Apply to Oregon or the draft lottery

A significant IV coefficient answers none of these questions by itself.

Practice Bridge

- Homework: `homework/session_04_homework.md`.
- R starter: `r_exercises/week_04_iv_late.R`.
- Self-check: `assessments/qcm/session_04_qcm.md`.

Student deliverable

Report the offer effect, the take-up effect and the local treatment effect.
Then draw one plausible threat and explain the affected assumption.

Paths start at the course root. Workload: approximately 60–75 minutes.

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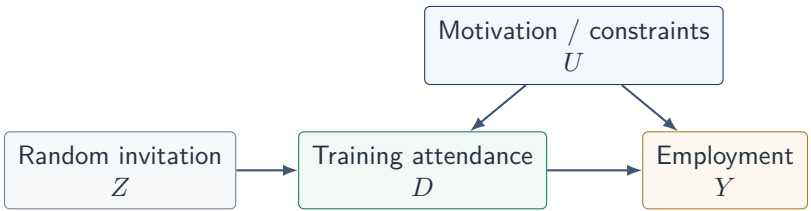
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From Our DAG to the Regression Equation

For this illustration, suppose everyone has the same treatment effect τ.



$$Y_i = \alpha + \tau D_i + u_i, \qquad \mathrm{Cov}(D_i, u_i) \neq 0.$$

The familiar endogeneity problem

u_i gathers other determinants of employment, including unobserved motivation. Attendance is related to these determinants, so its OLS slope need not be causal.

The IV Ratio: From Regression Back to Our Example

In the constant-effect model, a valid instrument satisfies:

$$\underbrace{\mathrm{Cov}(Z_i, D_i) \neq 0}_{\text{relevance}}, \qquad \underbrace{\mathrm{Cov}(Z_i, u_i) = 0}_{\text{exogeneity and exclusion}} .$$

$$\mathrm{Cov}(Z_i, Y_i) = \tau \, \mathrm{Cov}(Z_i, D_i) \quad \Rightarrow \quad \tau = \frac{\mathrm{Cov}(Z_i, Y_i)}{\mathrm{Cov}(Z_i, D_i)}.$$

For binary Z, this becomes the familiar Wald ratio:

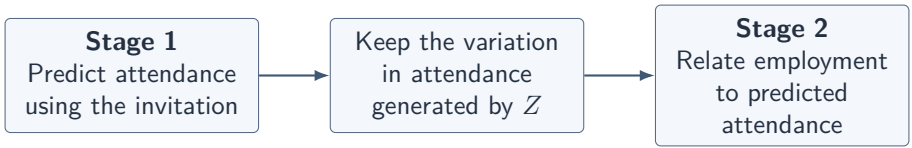
$$\frac{\mathrm{Cov}(Z_i, Y_i)}{\mathrm{Cov}(Z_i, D_i)} = \frac{ITT_Y}{ITT_D}.$$

Same observable ratio, a carefully defined target

Common effect: the ratio identifies τ. With heterogeneous effects of a binary treatment, the four LATE assumptions identify the **complier average**.

Two-Stage Least Squares: The Estimation Workflow

2SLS is the regression implementation of the IV idea.



- In the simple binary-instrument example, 2SLS gives the Wald estimate.
- The same method also accommodates continuous instruments or treatments.

Estimation and identification

Running 2SLS produces a coefficient. The instrument's causal argument makes it interpretable.

Two-Stage Least Squares: Equations and Interpretation

Stage 1: predict treatment

$$D_i = \pi_0 + \pi_1 Z_i + \eta_i$$

$$\widehat{D}_i = \widehat{\pi}_0 + \widehat{\pi}_1 Z_i$$

π_1 measures how Z moves D.
With binary Z, fitted attendance is the mean in each invitation arm.

Stage 2: use that prediction

$$Y_i = \alpha + \tau \widehat{D}_i + \varepsilon_i$$

In the invitation example:

$$\widehat{\tau}_{2SLS} = \widehat{\tau}_{Wald} = 0.20.$$

Under the LATE assumptions, this estimates the complier effect.

Implementation

Include the same pre-treatment controls X in both stages when needed.
Use an IV/2SLS command for correct standard errors.

Exit Ticket: Explain the Result to the Mayor

Complete these statements

1. The invitation increased employment by _____ percentage points.
2. Under the IV assumptions, training increased employment by an estimated _____ points for _____.
3. In IV notation, the invitation is ____, attendance is ____, and π_1 is the _____.

Memory aid: IV uses instrument-generated treatment variation;
its interpretation requires a credible design and a clearly named population.

Transition to Session 5

- IV uses a credible instrument Z_i to shift treatment received D_i.
- Some policies have no individual-level instrument.
- They may still vary across places or institutions over time.
- The next design asks whether untreated groups reveal the treated group's missing counterfactual trend.

Bridge question

When treatment timing differs across units, can changes in untreated units stand in for the treated counterfactual trend?

References: Empirical Designs

- Finkelstein, A., et al. (2012). "The Oregon Health Insurance Experiment: Evidence from the First Year." *Quarterly Journal of Economics*, 127(3), 1057–1106. doi:10.1093/qje/qjs020.
- Angrist, J. D. (1990). "Lifetime Earnings and the Vietnam Era Draft Lottery: Evidence from Social Security Administrative Records." *American Economic Review*, 80(3), 313–336. Author-hosted article.

Appendix: Population Targets and Sample Estimates

$$ITT_Y = E[Y \mid Z = 1] - E[Y \mid Z = 0],$$
$$ITT_D = E[D \mid Z = 1] - E[D \mid Z = 0].$$

Under independence, exclusion, relevance and monotonicity:

$$LATE = \frac{ITT_Y}{ITT_D} = E[Y_i(1) - Y_i(0) \mid D_i(1) > D_i(0)].$$

$$\widehat{\tau}_{IV} = \frac{\bar{Y}_{Z=1} - \bar{Y}_{Z=0}}{\bar{D}_{Z=1} - \bar{D}_{Z=0}}.$$

Target versus estimate: the population ratio identifies LATE under the assumptions; the sample ratio estimates it with sampling uncertainty.

Appendix: Independence and Exclusion in Notation

Independence:

$$Z_i \perp \{Y_i(1), Y_i(0), D_i(1), D_i(0)\}.$$

Exclusion: if $Y_i(d, z)$ allows assignment to have a separate effect,

$$Y_i(d, 1) = Y_i(d, 0) \quad \text{for each } d.$$

- Independence concerns how the instrument is allocated.
- Exclusion concerns how it can affect outcomes.
- Some designs defend independence conditional on pre-treatment X_i.

TOT Is Not Automatically LATE

$$TOT = E[Y_i(1) - Y_i(0) \mid D_i = 1]$$

- TOT is the average effect for actual treatment recipients.
- LATE is the average effect for compliers, $D_i(1) > D_i(0)$.
- With two-sided non-compliance, actual recipients include always-takers as well as compliers.
- In one-sided non-compliance, a treatment-on-the-treated reading may be defensible for the induced recipients.

Interpretive rule: do not call a Wald estimate "the effect on the treated" unless the compliance structure justifies it.

Multiple Instruments

- Multiple instruments can improve precision.
- They may identify a weighted average of different complier groups.
- Overidentification tests do not prove validity.
- Interpretability matters as much as statistical convenience.

Example: distance, eligibility, and encouragement may each move different people into training, so the resulting IV estimand may mix margins.

Appendix: Further Reading on IV and LATE

- Imbens, G. W., and Angrist, J. D. (1994). "Identification and Estimation of Local Average Treatment Effects." *Econometrica*, 62(2), 467–475.
- Angrist, J. D., Imbens, G. W., and Rubin, D. B. (1996). "Identification of Causal Effects Using Instrumental Variables." *Journal of the American Statistical Association*, 91(434), 444–455.
- Angrist, J. D., and Pischke, J.-S. (2009). *Mostly Harmless Econometrics*, IV chapter.